

Chapter 1: Introduction

Takashi Tanaka

Abstract. *This chapter presents the basic formulation of stochastic optimal control problems and outlines their classifications.*

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1.1 Stochastic optimal control: Basic setup

Broadly, the term *control system* refers to an artificial decision-making agent (also called a *controller*) that regulates and governs the dynamical behavior of an external environment (also called the *plant*). *Control theory* is a branch of mathematics that studies the decision-making rules for the agent (often called the *control policy* or *control law*) that achieve desirable behavior of the plant. The plant can be modeled as a dynamical input-output system that receives a time series of control commands u_t as input and generates a time series y_t of measured signals as output. The control policy to be designed is a causal mapping γ_t that, at each time step, maps the history of measured outputs y_s , $s \leq t$ to the current control input u_t .

A *deterministic* plant is one in which the output y_t is a deterministic function of the past inputs u_s , $s \leq t$. In contrast, a *stochastic* plant specifies a probability distribution over possible outputs y_t , parameterized by the past inputs u_s , $s \leq t$. Similarly, we call a control policy γ *deterministic* if the control input u_t is a deterministic function of the observed measurements y_s , $s \leq t$. A policy that assigns u_t randomly according to a probability distribution is called a *stochastic* or *randomized* policy. A control system involving either a stochastic plant or a stochastic policy is referred to as a *stochastic control system*.

1.1.1 Notational remarks

In this lecture note, we aim to keep our notation as consistent as possible with that used in the existing literature on stochastic control theory. Unfortunately, the notational conventions in the stochastic control literature often differ from those of probability theory. To avoid potential confusion later, we first present the standard notation in probability theory and then summarize how our notation differs from it.

Notation in probability theory

A probability space $(\Omega, \mathcal{F}, \mathcal{P})$ consists of a sample space Ω , a σ -algebra $\mathcal{F}$ of subsets of Ω , and a probability measure P defined on $\mathcal{F}$.¹ For each $E \in \mathcal{F}$ (called an *event*), the probability $\mathcal{P}(E) \in [0, 1]$ is assigned. A topological space $\mathcal{X}$ equipped with its Borel σ -algebra $\mathcal{B}_{\mathcal{X}}$ is denoted by $(\mathcal{X}, \mathcal{B}_{\mathcal{X}})$.² An $(\mathcal{X}, \mathcal{B}_{\mathcal{X}})$ -valued *random variable* X is a function

$$X : (\Omega, \mathcal{F}) \rightarrow (\mathcal{X}, \mathcal{B}_{\mathcal{X}}) \quad (1.1)$$

such that $\{\omega \in \Omega : X(\omega) \in B\} \in \mathcal{F}$ holds for each $B \in \mathcal{B}_{\mathcal{X}}$. The probability measure p_X induced by X is defined by

$$p_X(B) = \mathcal{P}(\{\omega \in \Omega : X(\omega) \in B\}), \quad \forall B \in \mathcal{B}_{\mathcal{X}}. \quad (1.2)$$

In probability theory, random variables are usually denoted by uppercase letters such as X , whereas lowercase letters such as x denote elements $x \in \mathcal{X}$.

The integral of a measurable function $f : \mathcal{X} \rightarrow \mathbb{R}$ over $B \in \mathcal{B}_{\mathcal{X}}$ with respect to p_X is denoted by $\int_B f(x) p_X(dx)$. If $\mathcal{X}$ is a countable space and there exists a function $P_X : \mathcal{X} \rightarrow [0, 1]$ such that $\int_B f(x) p_X(dx) = \sum_{x \in B} f(x) P_X(x)$ for each $B \in \mathcal{B}_{\mathcal{X}}$, then P_X is called *probability mass function (PMF)*. If $\mathcal{X}$ is an uncountable space and there exists a function $P_X : \mathcal{X} \rightarrow \mathbb{R}$ such that $\int_B f(x) p_X(dx) = \int_B f(x) P_X(x) dx$ for each $B \in \mathcal{B}_{\mathcal{X}}$, then P_X is called *probability density function (PDF)*. The mathematical expectation is defined by $\mathbb{E}^{p_X} [f(X)] = \int_{\mathcal{X}} f(x) p_X(dx)$.

Borel spaces

A space $(\mathcal{X}, \mathcal{B}_{\mathcal{X}})$ is said to be a (*standard*) *Borel space* if it is metrizable and complete separable. The additional assumptions on metrizability and complete separability are required for the property of being a Borel space to be preserved when taking countable Cartesian product.

Proposition 1.1 [2, Proposition 7.13] *Let $\mathcal{X}_1, \mathcal{X}_2, \dots$ be a sequence of Borel spaces. Then, the Cartesian products $\prod_{i=1}^n \mathcal{X}_i$ and $\prod_{i=1}^{\infty} \mathcal{X}_i$ are Borel spaces, and their Borel σ -algebras coincide with the product σ -algebras:*

$$\mathcal{B}_{\prod_{i=1}^n \mathcal{X}_i} = \prod_{i=1}^n \mathcal{B}_{\mathcal{X}_i}, \quad \mathcal{B}_{\prod_{i=1}^{\infty} \mathcal{X}_i} = \prod_{i=1}^{\infty} \mathcal{B}_{\mathcal{X}_i}.$$

Let X and Y be random variables taking values in Borel sets $(\mathcal{X}, \mathcal{B}_{\mathcal{X}})$ and $(\mathcal{Y}, \mathcal{B}_{\mathcal{Y}})$, respectively. Then, the joint measure $p_{X,Y}$ is defined by

$$p_{X,Y}(B) = \mathcal{P} \{ \omega \in \Omega : (X(\omega), Y(\omega)) \in B \}, \quad \forall B \in \mathcal{B}_{\mathcal{X} \times \mathcal{Y}}. \quad (1.3)$$

¹For a more complete treatment of this subject, see, e.g., [1].

²A topological space $\mathcal{X}$ is a set in which the notion of open subsets is well-defined. The Borel σ -algebra $\mathcal{B}_{\mathcal{X}}$ is the smallest σ -algebra containing all open sets of $\mathcal{X}$.

Stochastic kernels (Markov kernels)

The concept of conditional probability distributions should be formally defined as it is used routinely in this lecture note. For example, both the state transition model and the observation model in discrete-time stochastic control systems are characterized by conditional probability distributions. A control policy, the object we wish to design after all, is also expressed as a conditional probability distribution. In what follows, we formally refer to conditional probability distributions as *stochastic kernels* or *Markov kernels*.

Let $\mathcal{X}$ and $\mathcal{Y}$ be topological spaces equipped with Borel σ -algebras $\mathcal{B}_{\mathcal{X}}$ and $\mathcal{B}_{\mathcal{Y}}$, respectively. A (Borel-measurable) stochastic kernel $p(dy|x)$ on $(\mathcal{Y}, \mathcal{B}_{\mathcal{Y}})$ given $(\mathcal{X}, \mathcal{B}_{\mathcal{X}})$ is a parameterized family of probability measures satisfying the following conditions:

- (i) For every $x \in \mathcal{X}$, $p(\cdot|x)$ is a probability measure on $(\mathcal{Y}, \mathcal{B}_{\mathcal{Y}})$;
- (ii) For every $B \in \mathcal{B}_{\mathcal{Y}}$, $p(B|\cdot)$ is a Borel-measurable function on $\mathcal{X}$.

The next proposition shows that a joint probability measure on a product of Borel spaces can be decomposed into a marginal and a stochastic kernel.

Proposition 1.2 [2, Propositions 7.27] *Let $\mathcal{X}$ and $\mathcal{Y}$ be Borel spaces, and let $p_{X,Y}$ be a joint probability measure on $\mathcal{X} \times \mathcal{Y}$. Then, there exists a stochastic kernel $p_{Y|X}$ on $\mathcal{Y}$ given $\mathcal{X}$ such that*

$$p_{X,Y}(B_X \times B_Y) = \int_{B_X} p_{Y|X}(B_Y|x) p_X(dx) \quad \forall B_X \in \mathcal{B}_{\mathcal{X}}, B_Y \in \mathcal{B}_{\mathcal{Y}} \quad (1.4)$$

where p_X is the marginal of $p_{X,Y}$ on $\mathcal{X}$ defined by

$$p_X(B_X) = p_{X,Y}(B_X, \mathcal{Y}) \quad \forall B_X \in \mathcal{B}_{\mathcal{X}}. \quad (1.5)$$

Conversely, one can construct a unique probability measure on the product space from a given probability measure and one or more stochastic kernels on Borel spaces.

Proposition 1.3 [2, Propositions 7.28] *Let $\mathcal{X}_1, \mathcal{X}_2, \dots$ be a sequence of Borel spaces, and define $\mathcal{Y}_n = \prod_{i=1}^n \mathcal{X}_i$ and $\mathcal{Y}_{\infty} = \prod_{i=1}^{\infty} \mathcal{X}_i$. Let p_{X_1} be a probability measure on $\mathcal{X}_1$, and $p_{X_{n+1}|Y_n}$ be a stochastic kernel on $\mathcal{X}_{n+1}$ given $\mathcal{Y}_n$. Then, for $n = 2, 3, \dots$, there exist unique probability measures p_{Y_n} on $\mathcal{Y}_n$ such that*

$$\begin{aligned} p_{Y_n}(B_{X_1} \times B_{X_2} \times \dots \times B_{X_n}) &= \int_{B_{X_1}} \dots \int_{B_{X_{n-1}}} p_{X_n|Y_{n-1}}(B_{X_n}|x_1, \dots, x_{n-1}) \\ &\quad \times p_{X_{n-1}|Y_{n-2}}(dx_{n-1}|x_1, \dots, x_{n-2}) \dots \\ &\quad \times p_{X_2|X_1}(dx_2|x_1) p_{X_1}(dx_1) \\ &\quad \forall B_{X_1} \in \mathcal{B}_{\mathcal{X}_1}, \dots, B_{X_n} \in \mathcal{B}_{\mathcal{X}_n}. \end{aligned} \quad (1.6)$$

Furthermore, there exists a unique probability measure $p_{Y_{\infty}}$ on $\mathcal{Y}_{\infty}$ such that, for each $n = 1, 2, \dots$, the marginal of $p_{Y_{\infty}}$ on $\mathcal{Y}_n$ is p_{Y_n} .

Notation in this lecture note

A notable (and potentially confusing) convention in the stochastic control literature is that random variables are often denoted by lowercase letters such as x . Therefore, whenever an “ x ” appears, the reader must infer from context whether it represents a random variable or an element $x \in \mathcal{X}$.

Probability measures, joint measures, and conditional measures are denoted by $p_x(dx)$, $p_{x,y}(d(x, y))$, and $p_{y|x}(dy|x)$, respectively. As in probability theory, these may be abbreviated to $p(dx)$, $p(d(x, y))$, and $p(dy|x)$ when the meaning is clear. The following symbols will be used for particularly important conditional probability measures:

- τ for state-transition models;
- o for observation models;
- μ for control policies.

These concepts will be elaborated in the next subsection. Meanwhile, the symbol “ p ” is reserved for generic probability measures.

1.1.2 State space model

In this lecture note, we adopt the *state-space model* as a central framework for analyzing stochastic control systems. In a state-space model, the time-varying configuration of the plant is represented by a time-varying *state vector* x_t . Whenever we say that x_t is the state of the plant, we mean that it is a collection of variables that summarizes the plant’s history in a manner sufficient to predict its future behavior. More formally, the state random process $\{x_t\}$ is chosen to satisfy the *Markov property*, defined as follows:

Definition 1.1 A discrete-time random process $\{x_t\}$ is called a Markov process if

$$p(dx_{t+1} | x_t, x_{t-1}, x_{t-2}, \dots) = p(dx_{t+1} | x_t).$$

If the evolution of $\{x_t\}$ also depends on an auxiliary control input process $\{u_t\}$ such that

$$p(dx_{t+1} | x_t, x_{t-1}, x_{t-2}, \dots, u_t, u_{t-1}, u_{t-2}, \dots) = p(dx_{t+1} | x_t, u_t),$$

then $\{x_t\}$ is called a controlled Markov process.

Definition 1.2 The pair of random processes $\{x_t, y_t\}$ is called the hidden Markov process if

- $\{x_t\}$ is a Markov process;
- $\{y_t\}$ is directly observable, but $\{x_t\}$ is not;
- $p(dy_t | x_t, x_{t-1}, x_{t-2}, \dots) = p(dy_t | x_t)$.

The state-space model is essentially the “controlled hidden Markov process,” characterized by the following two components:

- **State transition kernel** $\tau_t(dx_{t+1}|x_t, u_t)$: This conditional probability distribution specifies the likelihood of a particular realization of the next state x_{t+1} , given the current state x_t and the current control input u_t . The state transition model $\tau_t(dx_{t+1}|x_t, u_t)$ includes the deterministic transition model $x_{t+1} = F_t(x_t, u_t)$ as a special case.
- **Observation kernel** $o_t(dy_t|x_t)$: This conditional distribution specifies how the current measurement y_t depends on the current state x_t . This formulation implicitly assumes that the state x_t cannot be directly measured by the controller, except in the special case where $y_t = x_t$. If $y_t = x_t$, the plant is said to be *fully observable*; otherwise, it is said to be *partially observable*.

Remark 1.1 *If both the state transition kernel and the observation kernel do not explicitly depend on time – that is, if $\tau_t(dx_{t+1}|x_t, u_t) = \tau(dx_{t+1}|x_t, u_t)$ and $o_t(dy_t|x_t) = o(dy_t|x_t)$ for all t – then the state space model is said to be time-invariant. Otherwise, the state space model is said to be time-varying.*

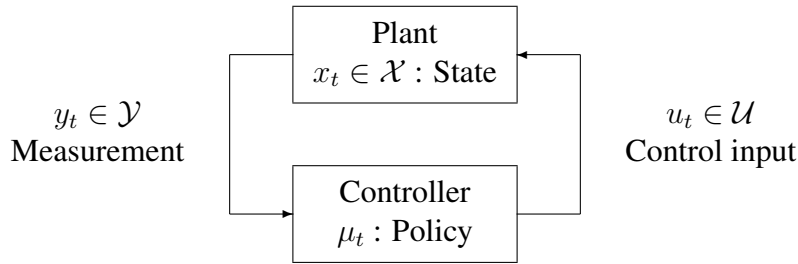


Figure 1.1: State space model.

Example 1.1 (*Linear-Gaussian model*) One of the most fundamental state-space models that we will frequently revisit in later sections is the linear-Gaussian model. In this model, the state transition probability $\tau_t(x_{t+1}|x_t, u_t)$ and the measurement probability $o_t(y_t|x_t, u_t)$ are given by

$$x_{t+1} = A_t x_t + B_t u_t + w_t, \quad w_t \sim \mathcal{N}(0, W_t) \quad (1.7)$$

$$y_t = C_t x_t + v_t, \quad v_t \sim \mathcal{N}(0, V_t). \quad (1.8)$$

In this model, the state space, control space, and measurement space are all Euclidean (i.e., $\mathcal{X} = \mathbb{R}^n$, $\mathcal{U} = \mathbb{R}^m$, and $\mathcal{Y} = \mathbb{R}^p$), and A_t , B_t , and C_t are matrices of appropriate dimensions. The random processes w_t and v_t are called the *process noise* and *measurement noise*, respectively.

Example 1.2 (*Finite-state system*) We will also frequently consider state-space models in which $\mathcal{X}$, $\mathcal{U}$, and $\mathcal{Y}$ are all finite sets. A finite-state model is a canonical setup for Markov Decision Processes (MDPs), Partially Observable Markov Decision Processes (POMDPs), and (tabular) reinforcement learning problems.

1.1.3 Classification of control policies

In the most general setting, the control policy to be designed is represented by a stochastic kernel of the form

$$\mu_t(du_t | y_t, y_{t-1}, \dots, u_{t-1}, u_{t-2}, \dots), \quad t = 0, 1, 2, \dots,$$

which assigns a probability mass or density to each possible control action u_t , given the information set $\mathcal{I}_t = \{y_t, y_{t-1}, \dots, u_{t-1}, u_{t-2}, \dots\}$ available to the decision-making agent at time t .

Deterministic vs. randomized policies

Frequently, it is sufficient to restrict attention to *deterministic policies*, where u_t is a deterministic function of the information set, i.e., $u_t = k_t(y_t, y_{t-1}, \dots, u_{t-1}, u_{t-2}, \dots)$ for some function $k_t(\cdot)$. If deterministic policies are used at all time steps, the sequence $(u_{t-1}, u_{t-2}, \dots)$ can be reconstructed from knowledge of $(y_t, y_{t-1}, \dots)$. Therefore, a deterministic policy can also be written as $u_t = k_t(y_t, y_{t-1}, \dots)$ without loss of generality.

A policy that is not deterministic is called a *randomized policy*. In general, the information set $\mathcal{I}_t$ is not equivalent to $\{y_t, y_{t-1}, \dots\}$ under randomized policies, and the best performance attainable by policies of the form $\mu_t(du_t | y_t, y_{t-1}, \dots, u_{t-1}, u_{t-2}, \dots)$ is at least as good as that attainable by policies of the form $\mu_t(du_t | y_t, y_{t-1}, \dots)$.

Static vs. dynamic policies

If the control input u_t depends only on the most recent measurement y_t , i.e.,

$$\mu_t(du_t | y_t, y_{t-1}, \dots, u_{t-1}, u_{t-2}, \dots) = \mu_t(du_t | y_t),$$

the policy is called *Markov* or *static*. A policy that is not static is called *dynamic*. Dynamic controllers generally achieve better performance than static controllers, but they require memory of the past. Static controllers, by contrast, have the advantage of simplicity since they do not require any memory.

State feedback vs. output feedback policies

Policies of the form $\mu_t(du_t | x_t, x_{t-1}, \dots, u_{t-1}, u_{t-2}, \dots)$ are admissible if the plant is fully observable (i.e., $y_t = x_t$). Such a policy is called *state feedback*. In contrast, general policies of the form $\mu_t(du_t | y_t, y_{t-1}, \dots, u_{t-1}, u_{t-2}, \dots)$ are called *output feedback*. State feedback offers substantial theoretical and practical advantages.

Choice of controller classes

In practice, policy randomization is often necessary in adaptive control and reinforcement learning algorithms [3]. It is also used in deceptive control and minimum-information control [4]. We will later study the class of *KL control problems* (Chapter 7), in which the optimal control laws naturally take the form of randomized policies.

Stochastic optimal control theory provides general results on how static and dynamic policies should be combined with state- and output-feedback configurations. When full state measurements are available, it can be shown that the best static state-feedback law performs as well as the

best dynamic state-feedback policy in a fairly broad class of performance metrics (see Chapter 3). Consequently, the category of “dynamic state feedback” is of limited relevance from an optimality standpoint. In the output-feedback setting, however, there is typically a gap between the performance attainable by static and dynamic policies, motivating careful study of the “dynamic output feedback” category. Nonetheless, “static output feedback” remains practically valuable, especially when simplicity is prioritized over performance.

Interestingly, despite the simplicity of the resulting control law, designing the best static output-feedback policy is often computationally more challenging than designing the best dynamic output-feedback policy. See Exercise 1.1 below.

Table 1.1 summarizes the common terminology used in the control theory literature.

	Fully observable	Partially observable
Markov policy	Static state feedback	Static output feedback
General policy	Dynamic state feedback	Dynamic output feedback

Table 1.1: Classification of feedback control policies

Exercise 1.1 (1) Consider the fully observable linear state-space model $x_{t+1} = Ax_t + Bu_t$, where (A, B) is a stabilizable pair. Provide an algorithm to compute K such that the static state feedback controller $u_t = Kx_t$ that stabilizes the closed-loop system.

(2) Consider the partially observable linear state-space model:

$$x_{t+1} = Ax_t + Bu_t \quad (1.9a)$$

$$y_t = Cx_t \quad (1.9b)$$

where (A, B) is a stabilizable pair and (A, C) is a detectable pair. Provide an algorithm to construct a dynamic output feedback controller that stabilizes the closed-loop system. Hint: Consider the observer-based controller — see, e.g., [5].

(3) Consider the state-space model (1.9) and suppose that our design goal is to design a static output feedback policy $u_t = Ky_t$ such that the closed-loop system is stable. What do we know about such a problems? See, e.g., [6, 7].

Exercise 1.2 Categorize the following exemplary control laws according to whether they are state or output feedback, static or dynamic feedback, and deterministic or randomized feedback. Complete Table 1.2.

- (i) *Linear Quadratic Regulator (LQR)*: We will study this in Chapter 3.
- (ii) *Linear Quadratic Gaussian (LQG)* controller: We will study this in Chapter 3.
- (iii) *P-control*: Consider the proportional control law $K(s) = K_P$ with a fixed reference signal r as shown in Figure 1.2.
- (iv) *PI-control*: Consider the proportional-integral control law $K(s) = K_P + \frac{K_I}{s}$ with a fixed reference signal r as shown in Figure 1.2.

- (v) *Q-learning*: Let $Q(x_t, u_t)$ be the Q-function, representing the cost of taking action u_t at state x_t . Let the Q-function be given.³ The soft-min policy (also known as the soft-max, depending on the sign of the objective function) is given by

$$\mu_t(u_t|x_t) = \frac{\exp\{-Q(x_t, u_t)/\lambda\}}{\sum_{u_t \in \mathcal{U}} \exp\{-Q(x_t, u_t)/\lambda\}}$$

where $\lambda > 0$ is called the temperature parameter.

- (vi) *Finite state controller*: Examine the structure of the policy described in [8, Section 2.2].

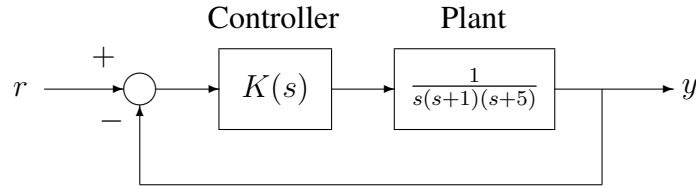


Figure 1.2: Classical feedback control system.

	State or output feedback	Static or dynamic	Deterministic or randomized
(i) LQR	State	Static	Deterministic
(ii) LQG control	Output	Dynamic	Deterministic
(iii) P-control			
(iv) PI-control			
(v) Q-learning			
(vi) Finite state controller			

Table 1.2: Examples of feedback control policies.

1.1.4 Cost Functional

In optimal control theory, the objective of the controller is specified by the cost functional. Choosing an appropriate cost functional not only introduces the notion of optimality but also enables a quantitative evaluation of a given control law's performance. The cost functionals we consider in the sequel are stage-additive. Specifically, a discrete-time finite-horizon stochastic optimal control problem is formulated as follows:

$$\min_{\mu_{0:T-1}} \mathbb{E}^{\mu_{0:T-1}} \left[\sum_{t=0}^{T-1} C_t(x_t, u_t) + C_T^{\text{exit}}(x_T) \right]. \quad (1.10)$$

³In Chapter 3, we study the *Q-learning* algorithm in which the Q-function is learned from historical data.

Here, $C_t(\cdot, \cdot)$ is the stage-additive instantaneous cost,⁴ and C_T^{exit} is the terminal cost.⁵ It is implicitly assumed that the minimization problem (1.10) is subject to dynamic constraints given by the state-space model $\tau_t(dx_{t+1}|x_t, u_t)$ and $o_t(dy_t|x_t)$ over the time steps $t = 0, 1, 2, \dots, T-1$. The minimization is over the space of causal policies. The symbol $\mu_{0:T-1}$ is a short-hand notation for the sequence of stochastic kernels

$$\mu_t(du_t|y_{0:t}, u_{0:t-1}), \quad t = 0, 1, \dots, T-1. \quad (1.11)$$

Once a policy (1.11) is fixed, the joint probability distribution of all random variables is defined as

$$p(dx_{0:T}, dy_{0:T-1}, du_{0:T-1}) = \prod_{t=0}^{T-1} \tau_t(dx_{t+1}|x_t, u_t) \mu_t(du_t|y_{0:t}, u_{0:t-1}) o_t(dy_t|x_t) p(dx_0). \quad (1.12)$$

The expectation $\mathbb{E}^{\mu_{0:T-1}}$ in (1.10) is taken with respect to the joint measure (1.12). The superscript of $\mathbb{E}^{\mu_{0:T-1}}$ emphasizes that this joint measure depends on the selected policy $\mu_{0:T-1}$.

1.2 Variations of the problem formulation

This section summarizes variations of the stochastic optimal control problems (1.10).

1.2.1 Full vs. partial observability

As we will see in subsequent sections, the difficulty of solving (1.10) crucially depends on whether the state process is fully observable (i.e., $y_t = x_t$) or partially observable (i.e., $y_t \neq x_t$). The stochastic optimal control problem (1.10) is commonly known as a *Markov Decision Processes (MDP)*, if $\mathcal{X}$ and $\mathcal{U}$ are finite spaces and the state x_t is fully observable. If $\mathcal{X}$, $\mathcal{Y}$ and $\mathcal{U}$ are finite, but the state x_t can only be observed through noisy measurements y_t , then (1.10) is called a *Partially Observable Markov Decision Processes (POMDP)*.

1.2.2 Finite vs. Infinite-Horizon Problems

As a natural extension of the finite-horizon problem (1.10), we are often interested in infinite-horizon stochastic optimal control problems, typically with the following assumptions:

- (i) The instantaneous cost function is time-invariant. That is, $C_t(x_t, u_t) = C(x_t, u_t)$ for $t = 0, 1, 2, \dots$.
- (ii) The state space model is time-invariant. That is, $\tau_t(dx_{t+1}|x_t, u_t) = \tau(dx_{t+1}|x_t, u_t)$ and $o_t(dy_t|x_t) = o(dy_t|x_t)$ for $t = 0, 1, 2, \dots$.

We note that formulations of infinite-horizon stochastic optimal control problems are sensitive to the order in which the expectation and the summation are taken. For example, $\min \mathbb{E} [\sum_{t=0}^{\infty} C(x_t, u_t)]$ and $\min \sum_{t=0}^{\infty} \mathbb{E} [C(x_t, u_t)]$ are different problems in general.

⁴One can also consider a slightly more general form $C_t(x_t, u_t, x_{t+1})$ without much additional complication.

⁵Conventionally, a stochastic optimal control problem is formulated as a cost minimization problem in control theory literature, whereas it is formulated as a reward maximization problem in reinforcement learning literature. In this note, we follow the former convention.

Exercise 1.3 Find an example random process $\{C_t\}$, $t = 0, 1, 2, \dots$ such that $\mathbb{E}[\sum_{t=0}^{\infty} C_t] \neq \sum_{t=0}^{\infty} \mathbb{E}[C_t]$.

The following variations of infinite-horizon problems have been extensively studied:

Variable end-time problems

We often encounter multistage decision-making problems in which the number of steps leading to the determination is not known in advance. Such a problem may be conveniently represented by an infinite-horizon variable end-time stochastic optimal control problem. To formulate such problems, one can introduce a subset $\mathcal{X}_{\text{abs}} \subset \mathcal{X}$ of absorbing states. An absorbing state can be used to recognize the end of an episode. Let $T_f = \min\{t > 0 | x_t \in \mathcal{X}_{\text{abs}}\}$ be the first exit time. In general, T_f is a random variable. The variable end-time problem is formulated as

$$\min_{\mu_{0:\infty}} \mathbb{E}^{\mu_{0:T_f-1}} \left[\sum_{t=0}^{T_f-1} C(x_t, u_t) + C^{\text{exit}}(x_{T_f}) \right]. \quad (1.13)$$

Discounted cost problems

Introducing a constant $\gamma \in (0, 1)$ called the *discount factor*, the infinite-horizon discounted-cost stochastic optimal control problem is formulated as:

$$\min_{\mu_{0:\infty}} \sum_{t=0}^{\infty} \gamma^t \mathbb{E}[C(x_t, u_t)]. \quad (1.14)$$

Notice that the discount factor places more weight on costs in the near future than on those in the distant future. One of the primary roles of the discount factor is to ensure that the infinite sum in (1.14) is bounded. The discount factor γ also plays a central role in guaranteeing the convergence of the *value iteration* algorithm (see Chapter 3), which serves as the foundation for various convergence results in reinforcement learning algorithms.

Average cost problems

Another commonly considered formulation of the infinite-horizon problem involves minimizing the average cost per unit time:

$$\min_{\mu_{0:\infty}} \limsup_{T \rightarrow \infty} \frac{1}{T} \sum_{t=0}^{T-1} \mathbb{E}[C(x_t, u_t)]. \quad (1.15)$$

In (1.13), (1.14) and (1.15), the minimization is over the space of causal policies $\mu_t(du_t | y_{0:t}, u_{0:t-1})$ for $t = 0, 1, 2, \dots$

1.2.3 Risk-sensitive control

Let us introduce the shorthand notation

$$C_{0:T} := \sum_{t=0}^{T-1} C_t(x_t, u_t) + C_T^{\text{exit}}(x_T),$$

and consider the following parametrized family of stochastic optimal control problems:

$$\min_{\mu_{0:T-1}} \mathcal{L}_\theta, \quad \theta \in \mathbb{R}, \quad (1.16)$$

$$\text{where } \mathcal{L}_\theta = \begin{cases} \frac{1}{\theta} \log \mathbb{E}[\exp(\theta C_{0:T})], & \text{if } \theta \neq 0, \\ \mathbb{E}[C_{0:T}], & \text{if } \theta = 0. \end{cases} \quad (1.17)$$

This class of problems is referred to as *risk-sensitive control*, and it contains our standard problem formulation (1.10) as a special case with $\theta = 0$.

The optimal policy obtained under $\theta = 0$ is called *risk-neutral*. A risk-neutral policy is optimized solely to minimize the expected cost $\mathbb{E}[C_{0:T}]$, and the variance $\text{Var}[C_{0:T}]$ does not enter the optimization criterion. Optimal policies obtained with $\theta < 0$ are called *risk-seeking*, in the sense that they tend to prefer higher variability in the cost. In contrast, $\theta > 0$ yields *risk-averse* policies that favor smaller cost variance.

Exercise 1.4 Consider the polynomial expansion of $\mathcal{L}_\theta$ in θ and derive the first two leading terms. Explain why varying $\theta \in \mathbb{R}$ controls the degree of risk sensitivity of the optimal policy.

The risk-sensitive control framework often allows cost volatility to be incorporated into the performance criterion without sacrificing computational tractability. For example, risk-neutral LQG problems can be generalized to the so-called *LEQG* (*Linear Exponential Quadratic Gaussian*) problems, which can be solved with only minor modifications to the standard LQG synthesis [9, 10].

1.2.4 Discrete vs. Continuous-Time Problems

The problem formulations introduced so far are called discrete-time stochastic optimal control problems, as the underlying state-space models operate in discrete time, and the control input u_t is determined at discrete time instances.

However, many physical processes are better described using continuous-time state-space models and decision policies that operate in continuous time. Moreover, continuous-time formulations sometimes offer more transparent and powerful policy synthesis methods. Despite these benefit, the discussion of continuous-time stochastic control problems requires the use of *stochastic differential equations* (*SDEs*), a subject with considerable mathematical depth in its own right. In this note, we will defer the discussion of continuous-time control problems until Chapter 5, following a brief introduction to continuous-time stochastic calculus in Chapter 4.

1.2.5 Model-Based, Simulator-Driven, and Data-Driven Syntheses

Policy synthesis algorithms can be categorized into different classes depending on what information about the problem is known to the decision-making agent.

Model-Based Synthesis: The most optimistic scenario is one in which all information about the problem (1.10), including analytical descriptions of the state-space model and the cost functional, is available. Policy design algorithms that leverage this prior knowledge are referred to as *model-based synthesis*.

Simulator-Driven Synthesis: In practice, there are many scenarios in which simulating a state space (i.e., sampling x_{t+1} from $\tau(dx_{t+1}|x_t, u_t)$ and y_t from $o(dy_t|x_t)$) is straightforward, but obtaining explicit analytical expressions for the conditional distributions $\tau(dx_{t+1}|x_t, u_t)$ and $o(dy_t|x_t)$ is difficult. Similarly, there are cases where evaluating the cost $C_t(x_t, u_t)$ for a particular realization of (x_t, u_t) is easy, but deriving an explicit expression for the function $C_t(\cdot, \cdot)$ is challenging.⁶ In such situations, a simulator can be used as a computational oracle from which the necessary information for policy synthesis is extracted. This approach is referred to as *simulator-driven synthesis*.

Data-Driven Synthesis: When neither an analytical model nor a simulator is available, the decision-making agent must learn about the plant solely from data obtained through interaction with it. In this case, the control policy must be constructed directly from the data, a process known as *data-driven policy synthesis*. Data-driven synthesis methods can be broadly categorized into *indirect* and *direct* approaches.

In *indirect methods*, the agent first attempts to construct a model of the plant based on observed data, a process commonly referred to as *system identification*. The resulting model is then used to synthesize the optimal policy using either model-based or simulator-driven techniques described above. In contrast, *direct methods* aim to derive the optimal policy directly from observed data, without constructing intermediate artifacts such as analytical models or simulators. Both approaches have been extensively studied in the adaptive control and reinforcement learning literatures.

While direct methods offer conceptual simplicity by eliminating the system identification step, indirect methods are often preferred in practice. One key reason is that prior knowledge about the physical laws governing the underlying plant dynamics (when it is available) can be incorporated during the system identification process. This approach is known as *gray-box modeling*. In contrast to *black-box modeling* which relies entirely on data, gray-box modeling leverages partial system knowledge, often enabling more accurate model construction with less data.

1.3 Organization of this lecture note

The rest of this lecture note is organized as follows. Chapter 2 introduces discrete-time stochastic optimal control problems. Based on Bellman's optimality principle, we develop an algorithmic foundation for model-based synthesis. Chapter 3 also addresses discrete-time stochastic optimal control problems, with an emphasis on simulator-driven and data-driven policy synthesis. In particular, we cover basic reinforcement learning algorithms for finite (PO)MDPs. Chapter 4 reviews selected topics in stochastic calculus that are required to formulate continuous-time stochastic

⁶For example, imagine a scenario where the function $C_t(\cdot, \cdot)$ is implemented as a trained neural network model.

optimal control problems in Chapter 5, where we also provide a formal derivation of the Hamilton–Jacobi–Bellman (HJB) equations.

Chapters 6, 7, and 8 focus on a special class of stochastic optimal control problems that are especially suitable for simulator-driven synthesis. Chapter 6 introduces the variational formula that serves as the foundation for these developments. Chapters 7 and 8 cover linearly solvable MDPs (for discrete-time settings) and path integral control (for continuous-time settings), respectively, and discuss how Monte Carlo simulations can be leveraged for effective policy synthesis.

Bibliography

- [1] R. B. Ash and C. A. Doléans-Dade, *Probability and measure theory*. Academic press, 2000.
- [2] D. Bertsekas and S. Shreve, *Stochastic Optimal Control: The Discrete-Time Case*. Athena Scientific, 1996.
- [3] R. S. Sutton and A. G. Barto, “Reinforcement learning: An introduction,” 2018.
- [4] T. Tanaka, P. M. Esfahani, and S. K. Mitter, “Lqg control with minimum directed information: Semidefinite programming approach,” *IEEE Transactions on Automatic Control*, vol. 63, no. 1, pp. 37–52, 2017.
- [5] K. Ogata, “Modern control engineering,” 2020.
- [6] M. Fu, “Pole placement via static output feedback is np-hard,” *IEEE Transactions on Automatic Control*, vol. 49, no. 5, pp. 855–857, 2004.
- [7] M. S. Sadabadi and D. Peaucelle, “From static output feedback to structured robust static output feedback: A survey,” *Annual reviews in control*, vol. 42, pp. 11–26, 2016.
- [8] N. Meuleau, K.-E. Kim, L. P. Kaelbling, and A. R. Cassandra, “Solving pomdps by searching the space of finite policies,” in *Proceedings of the Fifteenth conference on Uncertainty in artificial intelligence*, 1999, pp. 417–426.
- [9] P. Whittle, “Risk-sensitive linear/quadratic/gaussian control,” *Advances in Applied Probability*, vol. 13, no. 4, pp. 764–777, 1981.
- [10] D. Jacobson, “Optimal stochastic linear systems with exponential performance criteria and their relation to deterministic differential games,” *IEEE Transactions on Automatic control*, vol. 18, no. 2, pp. 124–131, 2003.